Supplementary material

Supplementary methods

Genetic variant analysis and selection

Genetic status was assessed using two complementary approaches. For clinical diagnostic purposes, genetic testing was performed in-house at Alzheimer Center Amsterdam, targeting one gene per request through Sanger sequencing or targeted next-generation sequencing panels. For research purposes, whole-exome sequencing (WES) was performed on DNA extracted from blood samples. Libraries were prepared using one of the following capture kits: Agilent_V6, Nimblegen_MedExome, Nimblegen_V3, or Twist_Comprehensive. WES was conducted on Illumina platforms. Reads were aligned to the GRCh38 reference genome supplemented with T2T patches [1] using DRAGMAP, and variants were called with DeepVariant. Single-nucleotide polymorphisms were subsequently processed in joint-calling mode with GLnexus, and resulting variants were annotated using ANNOVAR. Variants were selected based on likely pathogenic or pathogenic according to the American College of Medical Genetics and Genomics and the Association for Molecular Pathology (ACMG/AMP) guidelines [2] and DIAN classification [3]. *APP* duplications were included if at least the *APP* gene itself was duplicated.

Fluid collection and analysis

At clinical presentation, CSF was collected by lumbar puncture according to standard procedures and stored in the Amsterdam Dementia Biobank [4]. CSF Aβ42, p-tau181 and total tau were assessed with Innotest ELISA (Fujirebio) or Elecsys on Roche COBAS systems [5]. Amyloid positivity was confirmed using previously defined cut-off points [6,7]. Innotest values were converted to Elecsys-equivalent values using established transformation equations [8]. Additional CSF and plasma biomarkers were measured in stored material. CSF NfL concentrations were obtained from previously published data [9]. New CSF measurements, including NfL, GFAP, Aβ42, and Aβ40 (for the Aβ42/40 ratio), were performed using the Simoa® Neuro 4-plex E assay (Quanterix) on the Simoa HD-X platform. Plasma biomarkers p-tau217, NfL, GFAP, Aβ42, and Aβ40 were analyzed using the Simoa® ACC pTau217 (Janssen) assay and the Simoa® Neuro 4-plex E assay on the Simoa HD-X and results were harmonized across assay lots/batches to account for lot-related shifts in biomarker levels using Passing Bablok regression. For NfL and GFAP, age-adjusted z-scores were calculated using published formulas established in normal reference populations [9,10]. We additionally reported the corresponding percentile ranks for these z-scores.

Previously defined Youden's index thresholds for amyloid positivity were applied to plasma Aβ42/40, Aβ42, p-tau217, p-tau217/Aβ42, NfL, NfL age-adjusted, and GFAP. These thresholds had been harmonized to the lot number used to generate our current dataset by applying Passing Bablok regression formulas. The analyses included 795-803 Aβ- individuals and 937-944 Aβ+ individuals, depending on the marker because of missing data, and comprised cognitively unimpaired individuals, individuals with mild cognitive impairment (MCI) and individuals with different types of dementia [11]. The resulting thresholds were 0.050 for Aβ42/40, 6.1 pg/mL for Aβ42, 0.055 pg/mL for p-tau217, 0.0097 for p-tau217/Aβ42, 11.4 pg/mL for NfL, 0.36 (above 64th percentile) for NfL age-adjusted, and 66.3 for GFAP.

Measures of clinical characteristics and disease stage

We included six variables that reflect clinical characteristics related to disease stage: presence of symptoms, Mini-Mental State Examination (MMSE), Montreal Cognitive Assessment (MoCA), Medial Temporal Atrophy (MTA), Posterior Cortical Atrophy (PCA), and Estimated years to symptom onset (EYO). *Presence of symptoms* was defined as having a clinical diagnosis of MCI or dementia, compared to being tested cognitively normal. The *MMSE* [12] and *MoCA* [13] were administered by the attending physician as part of the diagnostic evaluation. When MoCA scores were missing, MMSE scores were converted to MoCA-equivalent scores [14]. Structural MRI scans were visually assessed for atrophy in key brain regions. Medial temporal lobe atrophy was rated using the *MTA* [15] scale, which ranges from 0 (no atrophy) to 4 (severe atrophy). Posterior cortical atrophy was rated using the *PCA* [16] scale, ranging from 0 (no atrophy) to 3 (knife-blade atrophy). For MTA and PCA, the left and right hemisphere scores were averaged. Additionally, global cortical atrophy score [17] and Fazekas score for white matter hyperintensities [18] were also assessed. All ratings were performed by trained research assistants in consensus with experienced neuroradiologists. *EYO* were calculated by subtracting the mutation age at onset (MAO) from the patient's age at clinical presentation (see MAO calculation and references below).

Measures of mutation characteristics

We included four variables to capture mutation characteristics: mutation-specific age at onset (MAO), variant-specific γ-secretase composite (GSC), age at diagnosis, and annual MMSE decline. MAO and GSC values were collected from data resources, while age at diagnosis and annual MMSE decline were collected during the clinical visits of our patients. Age at diagnosis and annual MMSE decline were included based on the hypothesis that more harmful mutations present symptoms earlier and are associated with faster cognitive decline.

The *MAO* for each genetic variant – i.e., the typical age at clinical symptom onset among carriers of a given pathogenic variant – was primarily derived from a recent comprehensive literature source (n = 12 of 15 included variants) [19]. For variants not reported in this source, MAO was estimated by integrating additional data from other published literature [20–22], including observed age at symptom onset data from cognitively impaired mutation carriers in our cohort, and clinical information from affected family members, derived from electronic medical records and family histories. A weighted MAO was then calculated for these variants by combining the available sources, with each contributing proportionally to their sample size of each source, to generate a robust MAO estimate (Supplementary Table 1). MAO was used as a proxy for mutation-associated phenotypic severity and should not be considered equivalent to pathogenicity classification under the ACMG/AMP framework [2] and subsequent ClinGen refinements, which assess whether a variant is disease-causing rather than how early or severely it manifests. MAO estimates may also be influenced by small sample sizes, family-specific genetic background, genetic modifiers, environmental factors, and variability in defining symptom onset.

Variant-specific *GSC scores* were derived from Schultz et al. [23], who quantified γ-secretase processivity in a HEK-cell assay using the ratio Aβ(37+38+40)/(42+43). Higher scores indicate greater processivity – i.e., preferential production of shorter Aβ species relative to longer species – and are reported on a 0-100 scale (0 = low, 100 = high γ-secretase activity). For variants missing from Schultz *et al.*, we cross-referenced a second source that reported Aβ37/42 [24], and harmonized those values to the primary 0-100 scale using a linear calibration model fitted on variants measured in both studies ($GSC_{Schultz}$ = -8.012 + GSC_{Lei} * 208.538, r = 0.56). For variants available in only one source (*PSEN1* p.A231V and *PSEN2* p.M239V), we imputed GSC values by applying this fitted model to predict the corresponding values on the Schultz 0-100 scale. For *APP*

p.M722K, only a single Aβ37/42 value was available [25]. We assumed this Aβ37/42 was comparable across sources and therefore benchmarked it against the overlapping *PSEN2* p.M239V Aβ37/42 value to impute the corresponding GSC score.

Age at diagnosis was defined as the age at clinical presentation with a diagnosis of MCI or dementia. Cognitively unimpaired individuals were excluded from this measure. *Annual MMSE decline* was estimated by calculating the change in MMSE score per year, by subtracting the last from the first MMSE score, divided by the time difference between these MMSE collections in years. Longitudinal MMSE data were available for 35 patients (59% of the total cohort), each of whom had more than one MMSE assessment. Of these, 13 patients had two assessments, 7 had three, and 15 had more than three assessments. The median follow-up duration was 1.6 years (interquartile range: 0 - 3.5 years).

Exploratory measure of variant pathogenicity

As an exploratory analysis, we additionally examined Rare Exome Variant Ensemble Learner (REVEL) scores as an established in silico measure of missense variant pathogenicity. REVEL integrates multiple individual prediction tools to estimate the likelihood that a missense variant is pathogenic, with scores ranging from 0 to 1 and higher values indicating greater predicted pathogenicity. REVEL scores were obtained from the Genomic Aggregation Database (gnomAD) for each included missense variant. REVEL was considered separately from the mutation-characteristic measures described above because it estimates whether a variant is disease-causing rather than the phenotypic severity or expected age at onset associated with that variant.

Statistical analysis

To assess whether plasma Aβ42/40 provided information on MAO independently of the other plasma biomarkers, an additional multivariable linear regression model included plasma Aβ42/40, p-tau217, age-adjusted GFAP, and age-adjusted NfL simultaneously. All predictors were standardized within the analytic sample. The incremental contribution of each biomarker was assessed by comparing the full model with reduced models omitting the respective biomarker and calculating the change in explained variance (ΔR^2) and partial R^2. Pairwise contrasts of the standardized regression coefficients were used to test whether the association of Aβ42/40 with mutation age at onset differed from those of the other plasma biomarkers.

Supplementary results

Genetic characteristics

59 patients carried 14 different pathogenic DIAD variants: 11 *PSEN1* variants, 1 *PSEN2* variant, and 2 *APP* variants, including different *APP* duplications ranging in size from 0.418 to 7.531 Mb (Supplementary Table 1). The most prevalent variants were *PSEN1* p.A79V ($n = 16$), *PSEN1* p.L262F ($n = 11$), and *APP* duplication ($n = 7$), while the remaining variants were observed in 1 to 5 patients each. One patient with a *PSEN1* c.338+1del mutation carried an additional variant of uncertain significance, i.e., *APP* c.-200G>A. The literature-derived MAO ranged between 35 and 63 years, and γ-secretase composite (GSC) scores ranged between 28 and 77. For two variants (*PSEN1* p.R377M and p.L418F), MAO data were not available from the DIAN cohort and were instead estimated using literature sources and data from our own cohort. GSC scores were unavailable for three missense variants (*PSEN1* p.L262F, p.R377M, and p.L418F).

Plasma biomarker characteristics

Among 50 patients with available plasma Aβ42/40, GFAP, and NfL measurements, 44 (88%) showed normal plasma Aβ42/40 ratios above the previously established amyloid positivity threshold. This was observed in both cognitively impaired (36/41, 88%) and cognitively unimpaired individuals (8/9, 89%). Only 3 of 7 *APP*/*APP* duplication carriers and 3 of 43 *PSEN1*/*PSEN2* carriers had ratios below this threshold (Supplementary Figure 1). The highest plasma Aβ42/40 ratios were observed in *PSEN1* p.G384A, *PSEN1* p.L262F, and *APP* p.M722K carriers (Supplementary Figure 2). Abnormal plasma p-tau217 concentrations were observed in 43 out of 47 individuals with available plasma (92%), comprising all 38 impaired patients and 5 of the 9 unimpaired patients. Abnormal p-tau217/Aβ42 ratios were observed in only 38 out of 47 individuals (81%; 36/38 impaired and 2/9 unimpaired). Median plasma GFAP age-adjusted z-scores corresponded to the 97.2nd percentile (range 13-100), and median plasma NfL age-adjusted z-scores corresponded to the 81.7th percentile (range 5-100). Elevated plasma GFAP concentrations above the amyloid positivity threshold were detected in 41 individuals (82%; 38/41 impaired and 3/9 unimpaired), and elevated plasma age-adjusted NfL z-scores exceeding the amyloid positivity threshold were detected in 38 individuals (76%; 35/41 impaired and 3/9 unimpaired).

Supplementary references

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Supplementary Table 1 Genetic and family characteristics of DIAD carriers with pathogenic variants identified in the cohort

Variant	Exon	Function	GMAF*	MAO†	REVEL	GSC	ACMG/AMP classification**	ACMG/AMP evidence criteria	First/early report	N total	N FHΦ
PSEN1											
p.A79V	4	Missense	9.29×10^{-6}	60	0.958	58.58	Likely pathogenic	PS3, PP1, PP2, PP3	Cruts et al., 1998	16	16 (1-7)
p.Y115C	5	Missense	$<1\times10^{-6}$	45	0.980	29.78	Pathogenic	PS3, PS4, PM1, PM2, PM5, PP2, PP3	Cruts et al., 1998	5	5 (2-7)
p.G206D	7	Missense	1.74×10^{-5}	37	0.922	52.47	Pathogenic	PS3, PM1, PM2, PM5, PP2, PP3	Raux et al., 2005	2	2 (1-3)
p.A231V	7	Missense	$<1\times10^{-6}$	56	0.902	77.49	Likely pathogenic	PM1, PM2, PP2, PP3	Cruts et al., 1998	2	2 (3-6)
p.L262F	8	Missense	$<1\times10^{-6}$	53	0.931	UNK	Likely pathogenic	PS3, PM1, PM2, PP2, PP3	Forsell et al., 1997	11	8 (2-6)
p.P264L	8	Missense	1.24×10^{-6}	49	0.965	33.74	Pathogenic	PS3, PM1, PM2, PM5, PP1, PP2, PP3	Campion et al., 1995	3	2 (1-8)
p.R269H	8	Missense	9.3×10^{-6}	58	0.945	51.14	Likely pathogenic	PS3, PM1, PP2, PP3, BS1	Gómez-Isla et al., 1997	2	2 (1-2)
p.R377M	10	Missense	$<1\times10^{-6}$	43	0.957	UNK	Likely pathogenic	PM1, PM2, PP2, PP3	Janssen et al., 2003	1	1 (1)
p.G384A	11	Missense	$<1\times10^{-6}$	35	0.959	27.73	Pathogenic	PS3, PM1, PM2, PP1, PP2, PP3	Cruts et al., 1998	4	3 (5-8)
p.L418F	12	Missense	$<1\times10^{-6}$	36	0.906	UNK	Likely pathogenic	PS2, PS3, PM1, PM2, PP2, PP3	Rogaeva et al., 2001	2	1 (3)
c.338+1del	-	Deletion	$<1\times10^{-6}$	43	-	-	Pathogenic	PS3, PM1, PM2, PP1	Tysoe et al., 1998	1	1 (6)
PSEN2											
p.M239V	8	Missense	$<1\times10^{-6}$	63	0.877	33.70	Pathogenic	PS3, PS4, PM1, PM2, PM5, PP3	Rogaev et al., 1995	2	2 (4-5)
APP											
Duplication	-	Copy number gain	$<1\times10^{-6}$	50	UNK	-	Pathogenic	ACMG/ClinGen CNV criteria	Rovelet-Lecrux et al., 2006	7	4 (1-7)
p.M722K	17	Missense	$<1\times10^{-6}$	48	UNK	33.70	Pathogenic	PS3, PM1, PM2, PP2, PP3	Wang et al., 2015	1	1 (3)

*Based on gnomAD v4.1.0 (Nov 2025). Variants not observed in gnomAD are reported as $<1\times10^{-6}$, calculated as 1/AN using a maximum AN of $\sim1.5\times10^{-6}$. AN denotes the total number of alleles with high-quality genotype calls at the variant site (≈2× the number of individuals). ** ACMG/AMP classifications were obtained from the Alzforum Mutations database and/or the DIAN-TU variant eligibility list, which incorporates ACMG/AMP guidelines (Richards *et al.*, 2015), published literature, variant databases, and evidence from DIAN studies. *PSEN1* p.R377M had conflicting results but was in line with clinical procedure included in the study. Supporting evidence criteria for sequence variants were obtained from the Alzforum Mutations database. *APP* p.M722K was classified according to published variant-specific evidence. Original or variant-specific reports are cited separately where available. Copy-number variants were considered according to the ACMG/ClinGen technical standards for CNV interpretation (Riggs et al., 2020). † Mutation age at onset (MAO) is based on Llibre-Guerra et al. (2024), and if unavailable, a weighted sum of the observed age at onset of patients in our cohort, their family history, and available other literature (Janssen et al., 2003, Leverenz et al. 2006, and Reinert et al. 2014). Φ Number of family histories (range of number of affected individuals in family history). Abbreviations: ACMG = American College of Medical Genetics and Genomics; AMP = Association for Molecular Pathology; CADD = Combined Annotation Dependent Depletion; CNV = copy number variant; FH = family history; GMAF = global minor allele frequency; p = protein; REVEL = Rare Exome Variant Ensemble Learner; UNK = unknown.

Supplementary Table 2 Multivariable associations of plasma biomarkers with mutation age at onset

A) Multivariable regression model

Predictor	β (years per SD)	SE	95% CI	*P*-value	Partial R^2	ΔR^2
Aβ42/40	-5.49	0.87	-7.24, -3.74	1.36E-07	0.488	0.352
p-tau217	-4.01	1.38	-6.79, -1.22	0.00585	0.167	0.074
GFAP, age-adjusted	0.88	1.35	-1.85, 3.60	0.519	0.01	0.004
NfL, age-adjusted	1.04	1.32	-1.63, 3.71	0.437	0.014	0.005

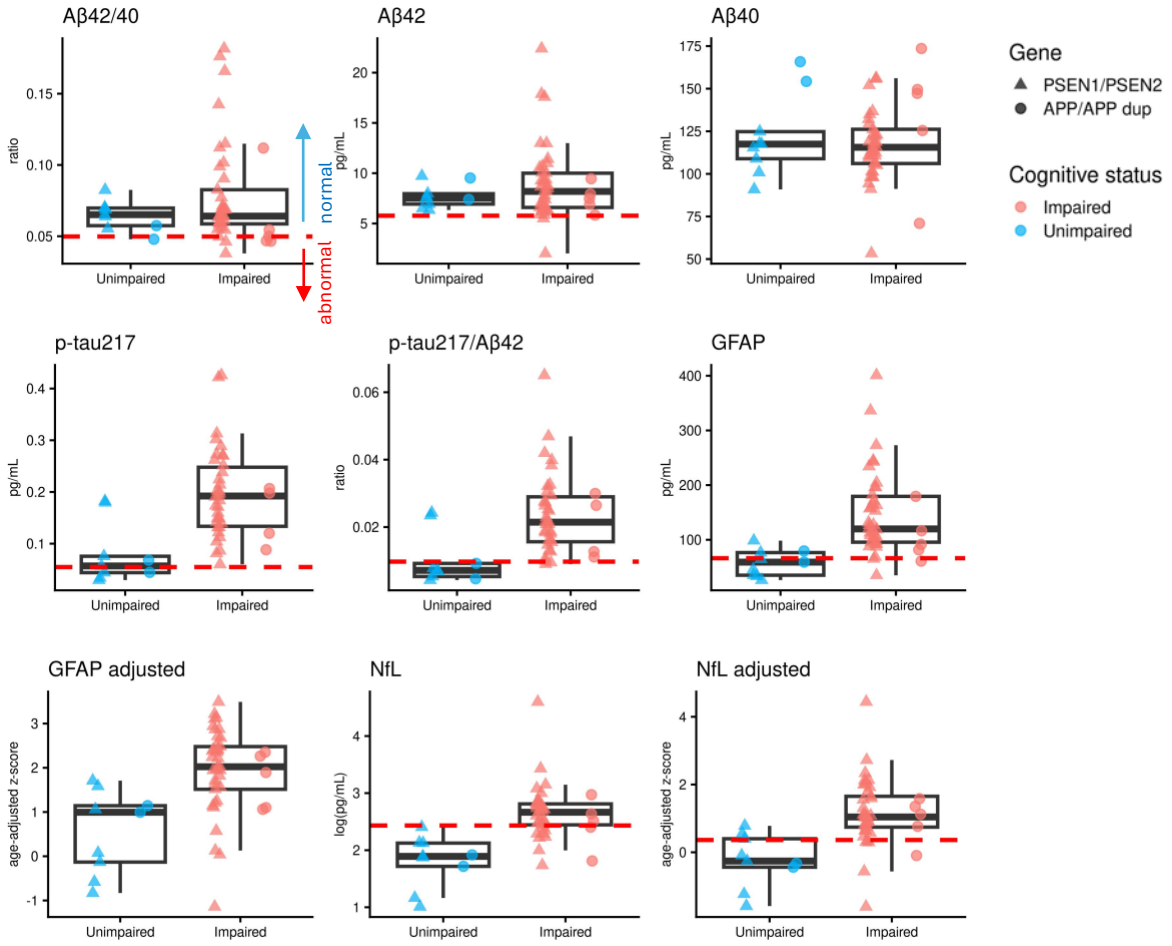
N = 47; model R^2 = 0.631; adjusted R^2 = 0.596.

B) Pairwise comparison of biomarker coefficients

Comparison	B difference	SE	*P*-value
Aβ42/40 vs p-tau217	-1.48	1.7	0.389
Aβ42/40 vs GFAP	-6.37	1.73	0.000659
Aβ42/40 vs NfL	-6.53	1.56	0.000147

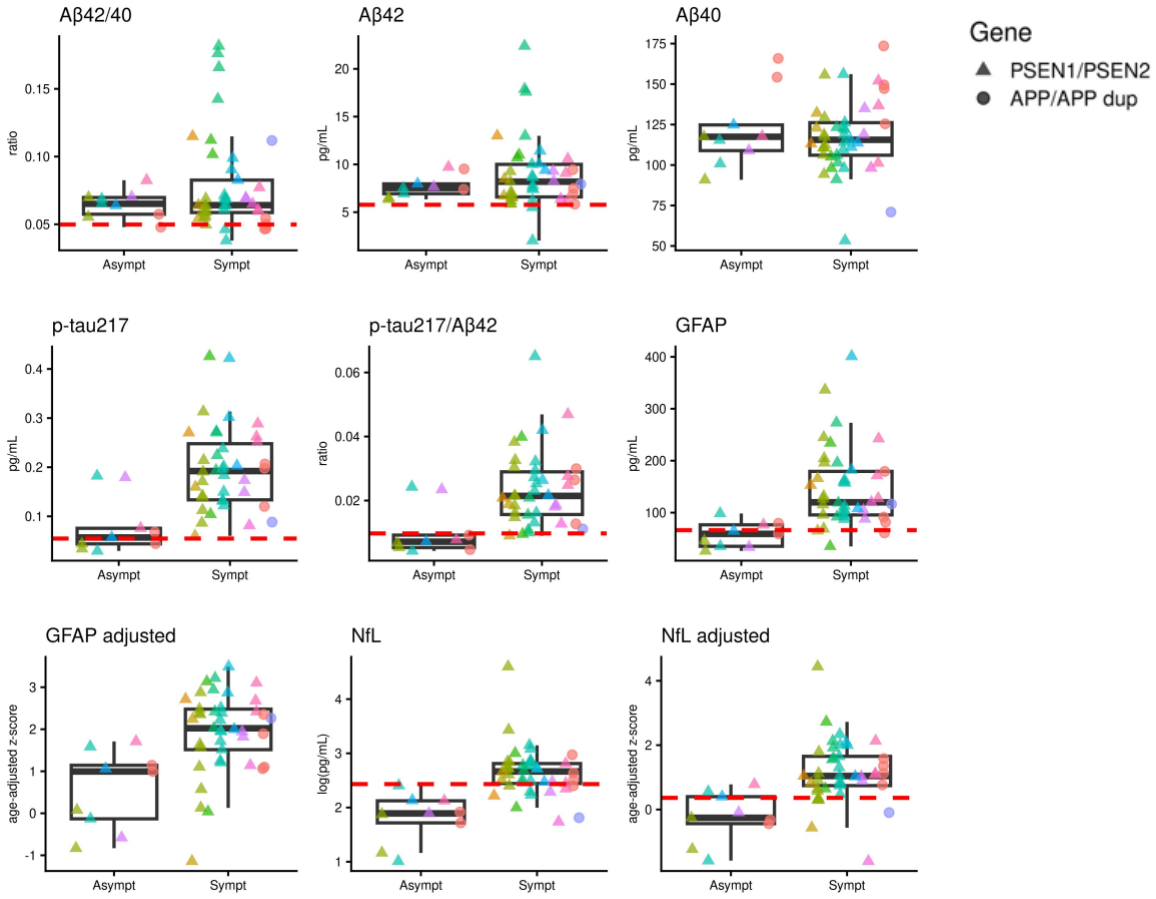
(A) Multivariable linear regression model of plasma biomarkers with mutation age at onset (MAO). Plasma Aβ42/40, p-tau217, age-adjusted GFAP, and age-adjusted NfL were included simultaneously in the model. All biomarker measures were standardized within the analytic sample before analysis. Regression coefficients (β) therefore represent the difference in MAO, in years, per 1-SD increase in the respective biomarker, adjusted for the other biomarkers in the model. Partial R^2 represents the proportion of residual variance in MAO uniquely explained by each biomarker conditional on the other biomarkers. ΔR^2 represents the increase in model R^2 when the respective biomarker was added to a model containing the other three biomarkers. The full model included 47 participants and explained 63.1% of the variance in MAO (R^2 = 0.631; adjusted R^2 = 0.596). (B) Pairwise comparisons of regression coefficients from the multivariable model. Differences in β represent the estimated difference between the standardized regression coefficients for plasma Aβ42/40 and each comparator biomarker. P-values test the null hypothesis that the two regression coefficients are equal. Abbreviations: SD = standard deviation; SE = standard error.

Supplementary Figure 1 Distribution of plasma biomarkers with amyloid-positivity thresholds



This figure depicts plasma biomarkers levels of cognitively unimpaired and impaired DIAD mutation carriers. The red lines show the threshold of amyloid positivity per marker. Abbreviations: CU = cognitively unimpaired; dup = duplication.

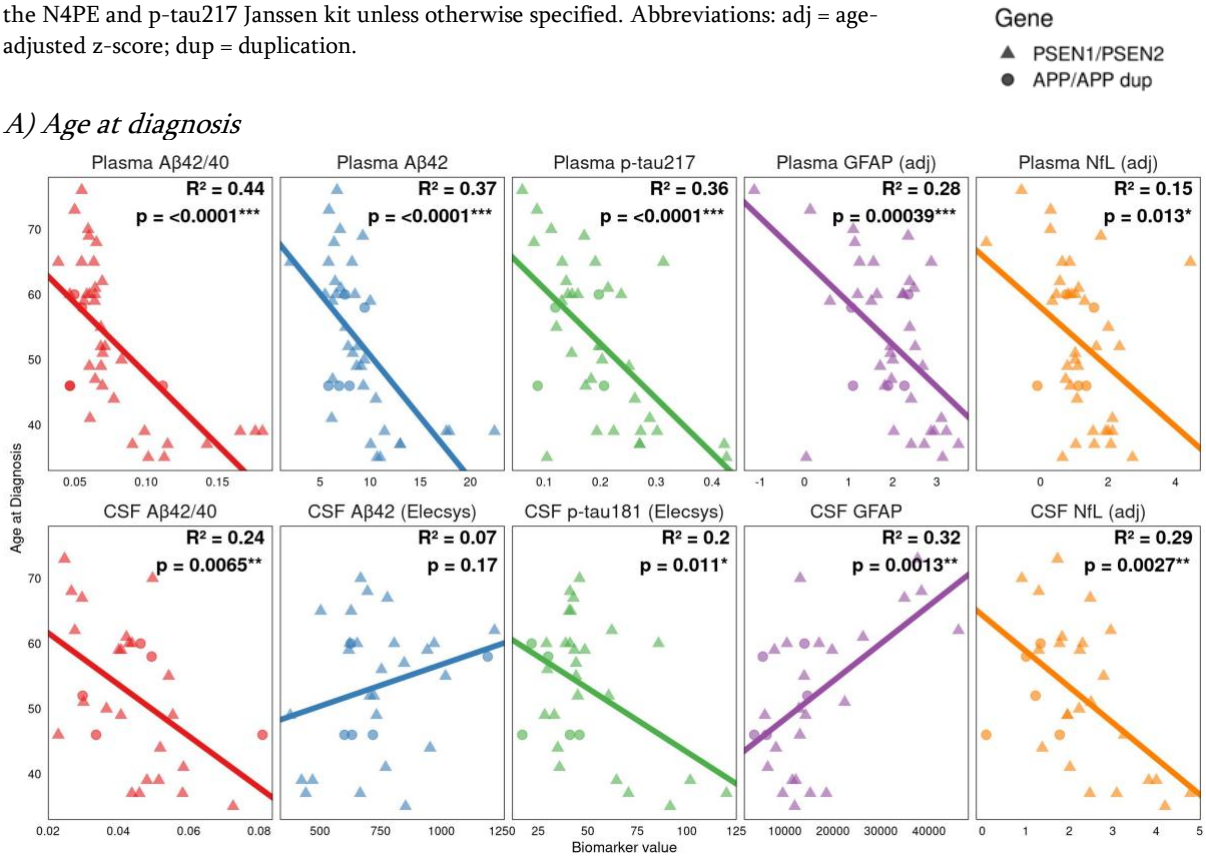
Supplementary Figure 2 Plasma biomarkers highlighted by DIAD mutation-specific levels



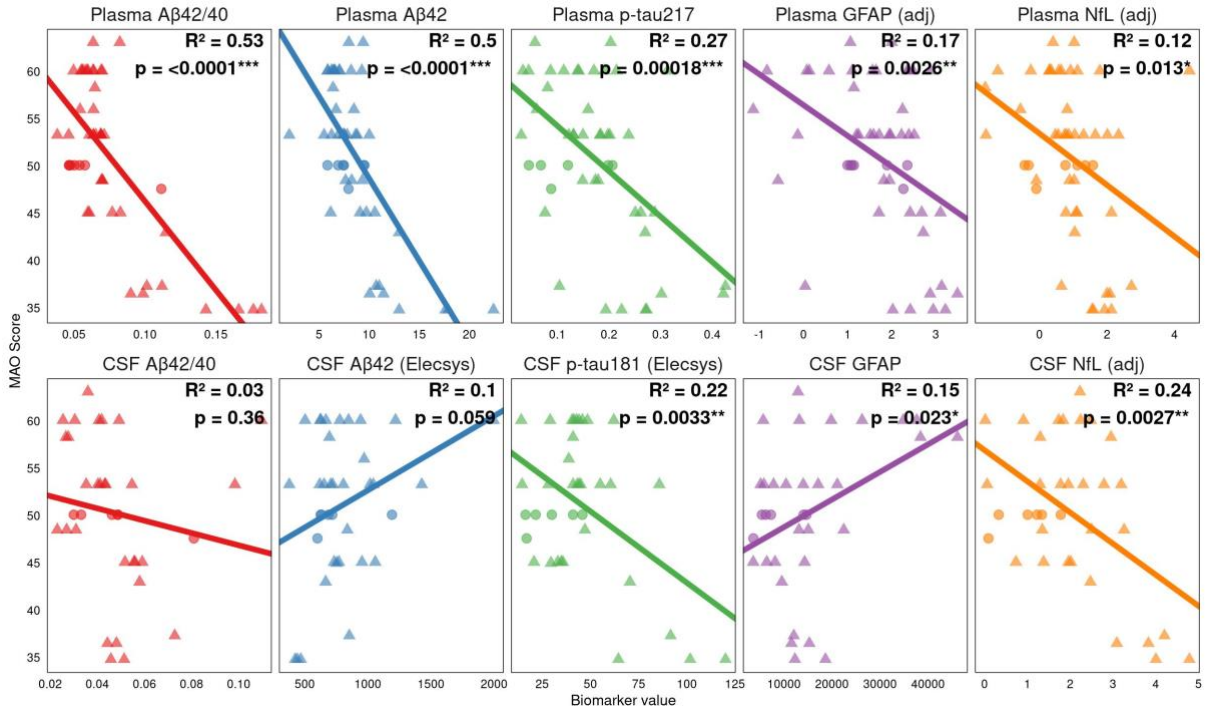
These figures show plasma biomarker levels stratified by cognitively unimpaired and impaired individuals. Colors represent different mutation groups (labels omitted to preserve participant anonymity). The red line shows the threshold of amyloid positivity. Variant names are not listed in the legend to protect participant privacy. Abbreviations: dup = duplication.

Supplementary Figure 3 Fluid biomarkers and their association with clinical and mutation characteristics

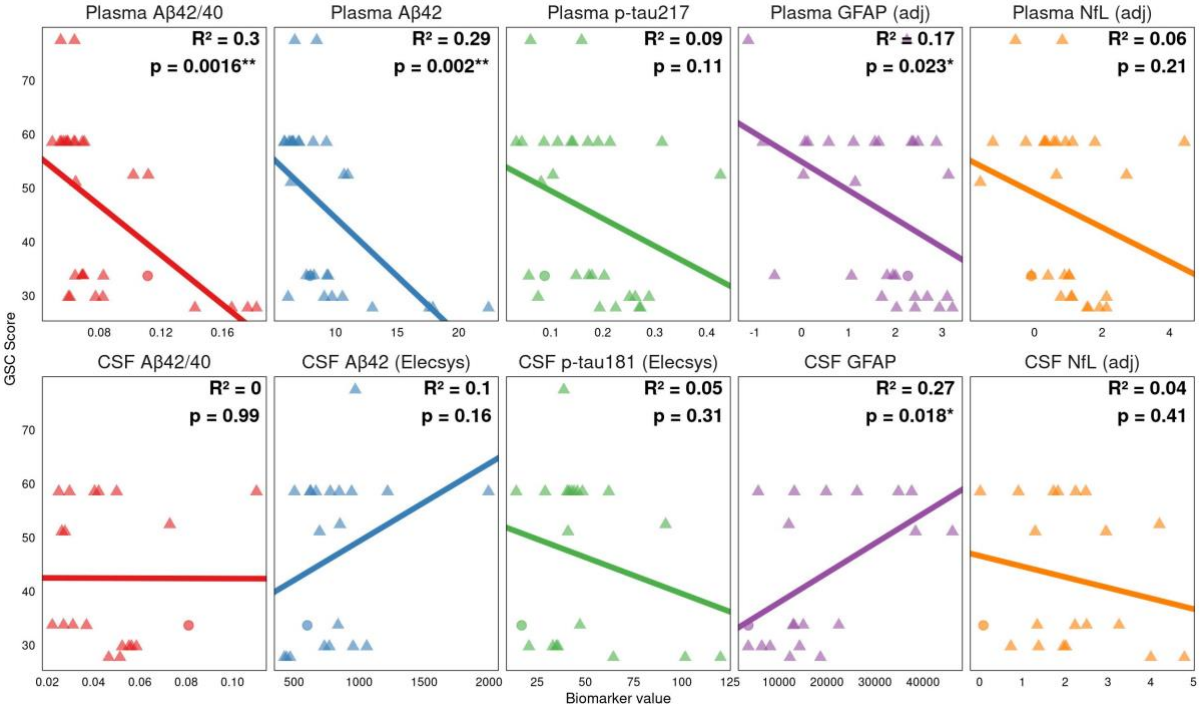
These figures show unadjusted associations between plasma and CSF biomarker levels and clinical and mutation-related characteristics, using linear and logistic regression models. Biomarkers are measured with the N4PE and p-tau217 Janssen kit unless otherwise specified. Abbreviations: adj = age-adjusted z-score; dup = duplication.



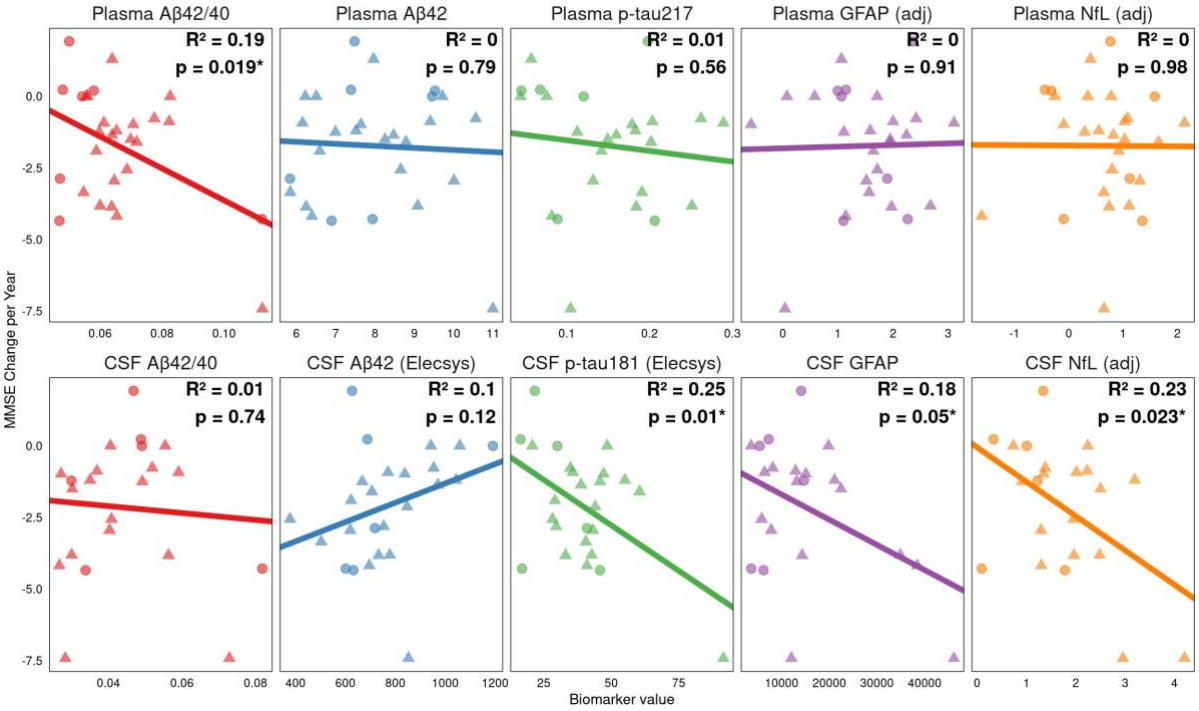
B) Mutation age at onset (MAO)



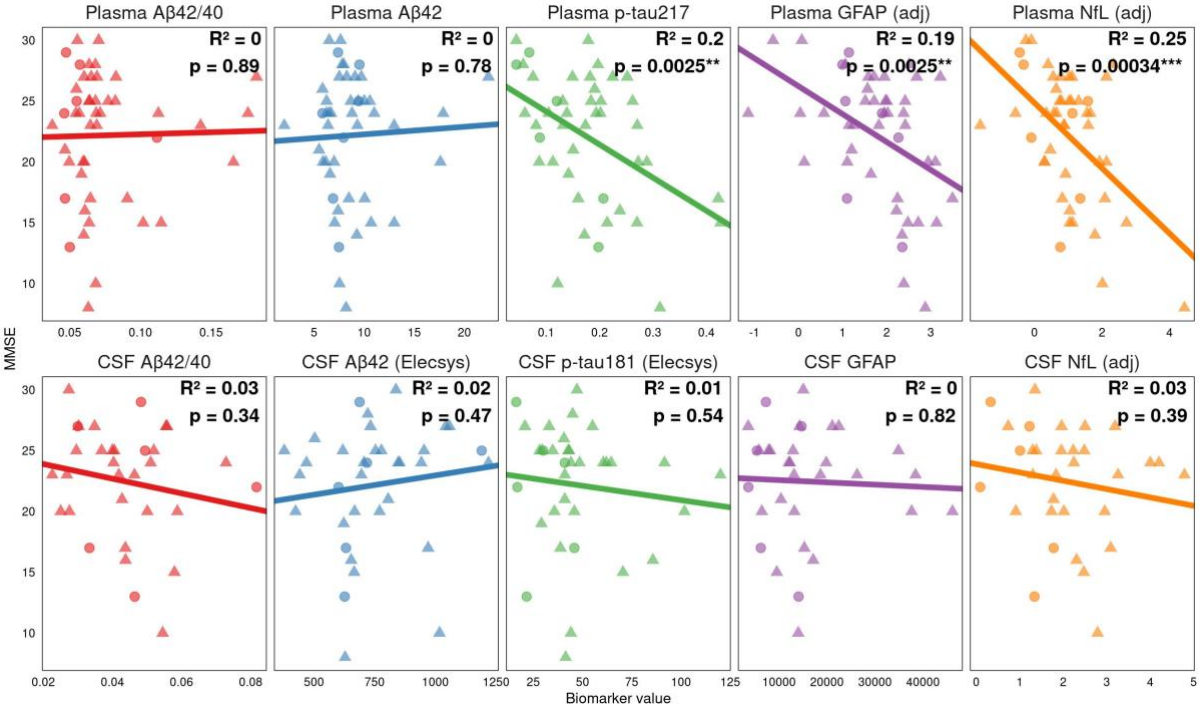
C) γ-secretase composite (GSC)



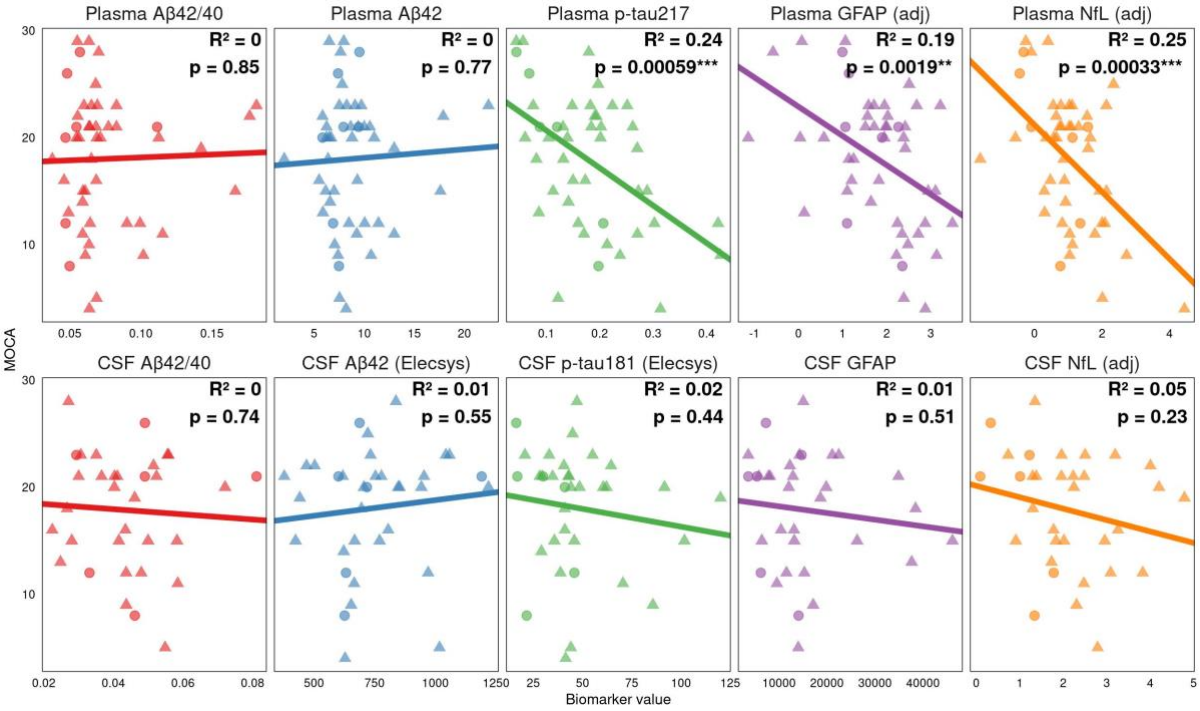
D) MMSE change per year



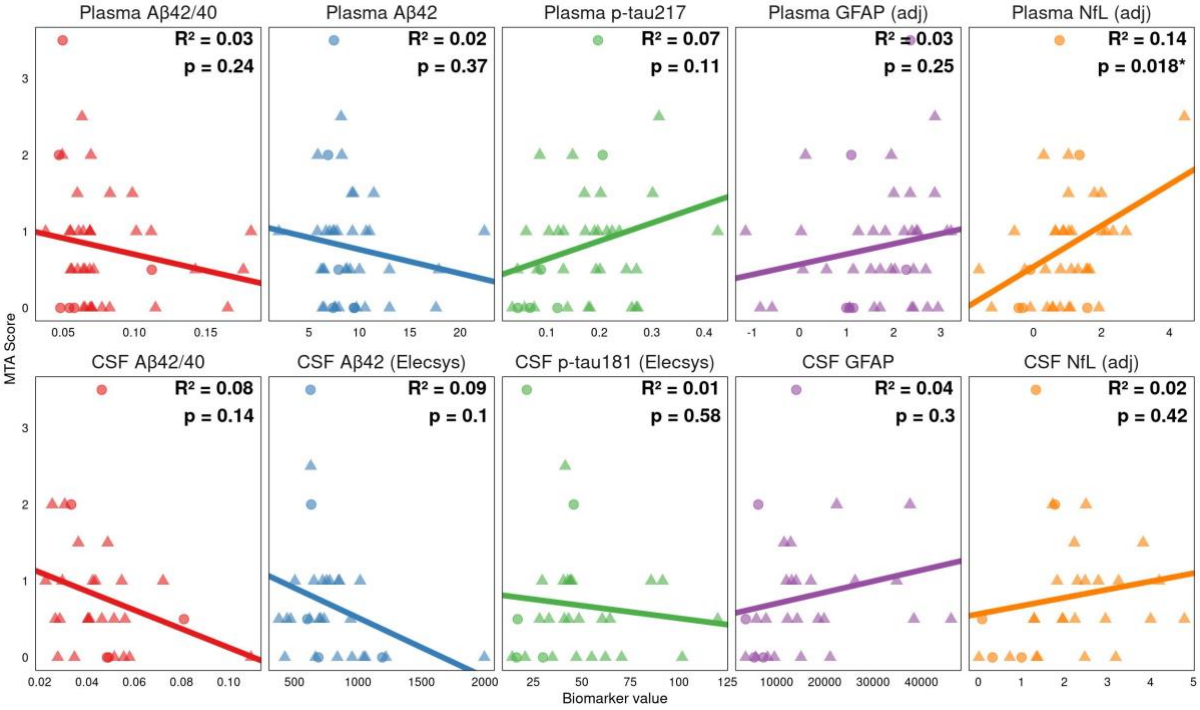
E) Mini-Mental State Examination (MMSE)



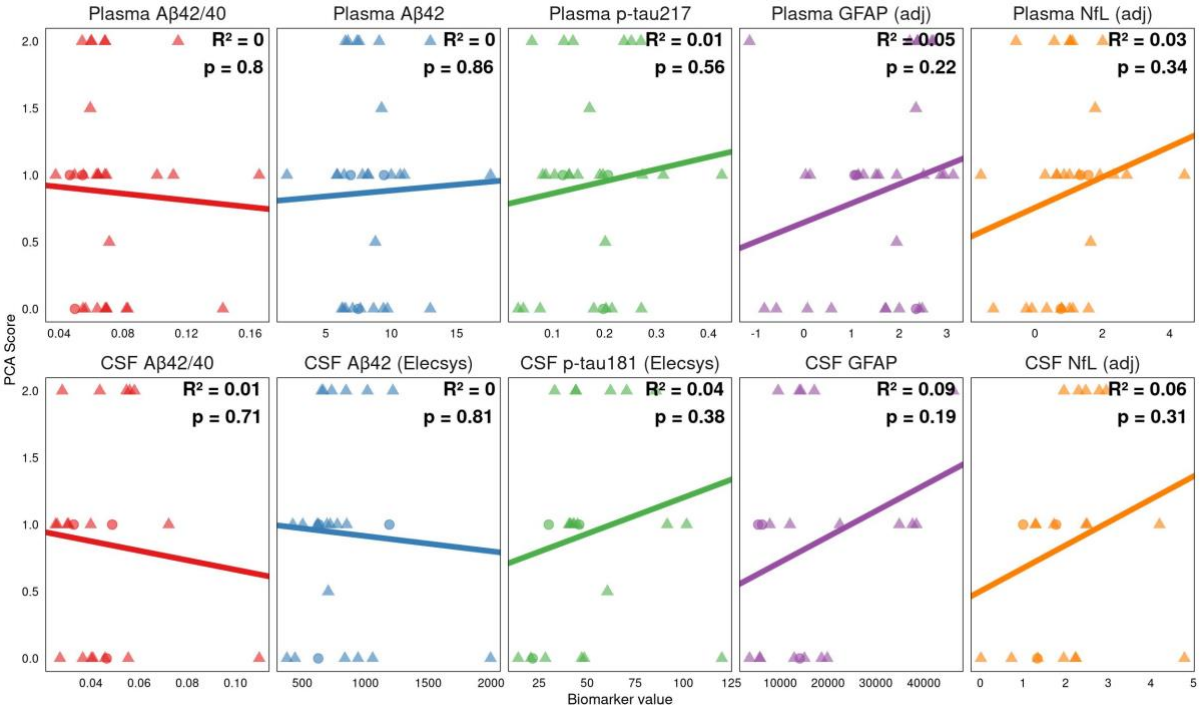
F) Montreal Cognitive Assessment (MoCA)



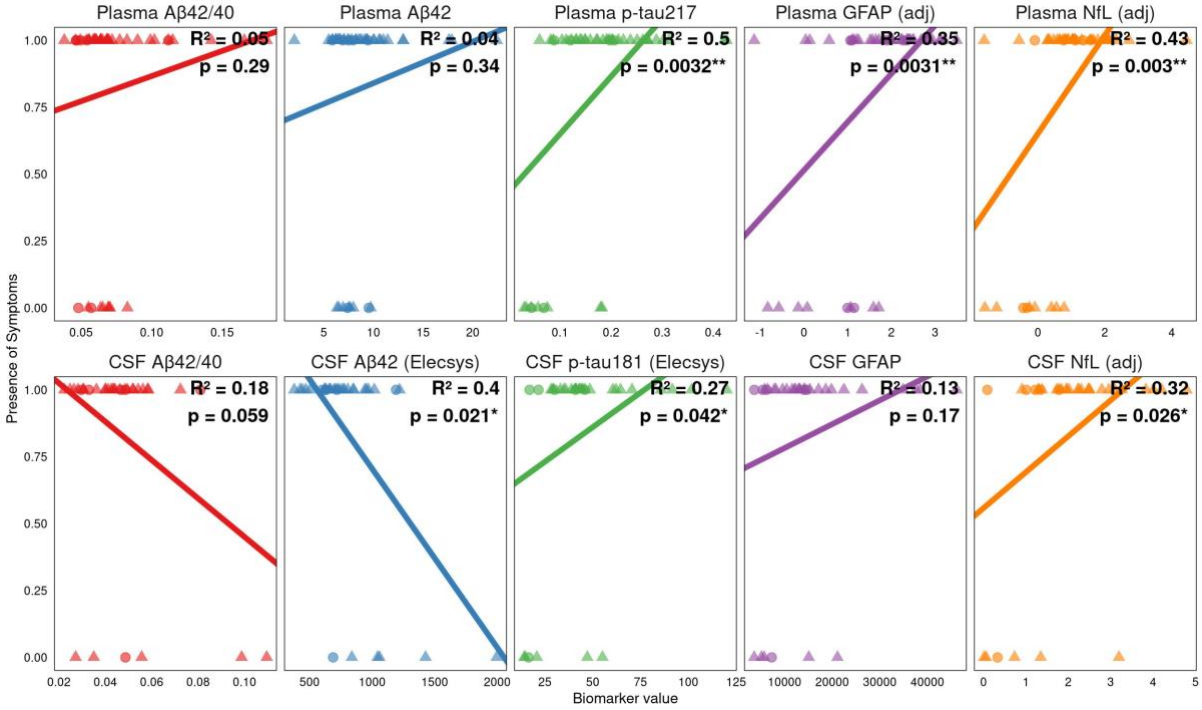
G) Medial temporal atrophy (MTA)



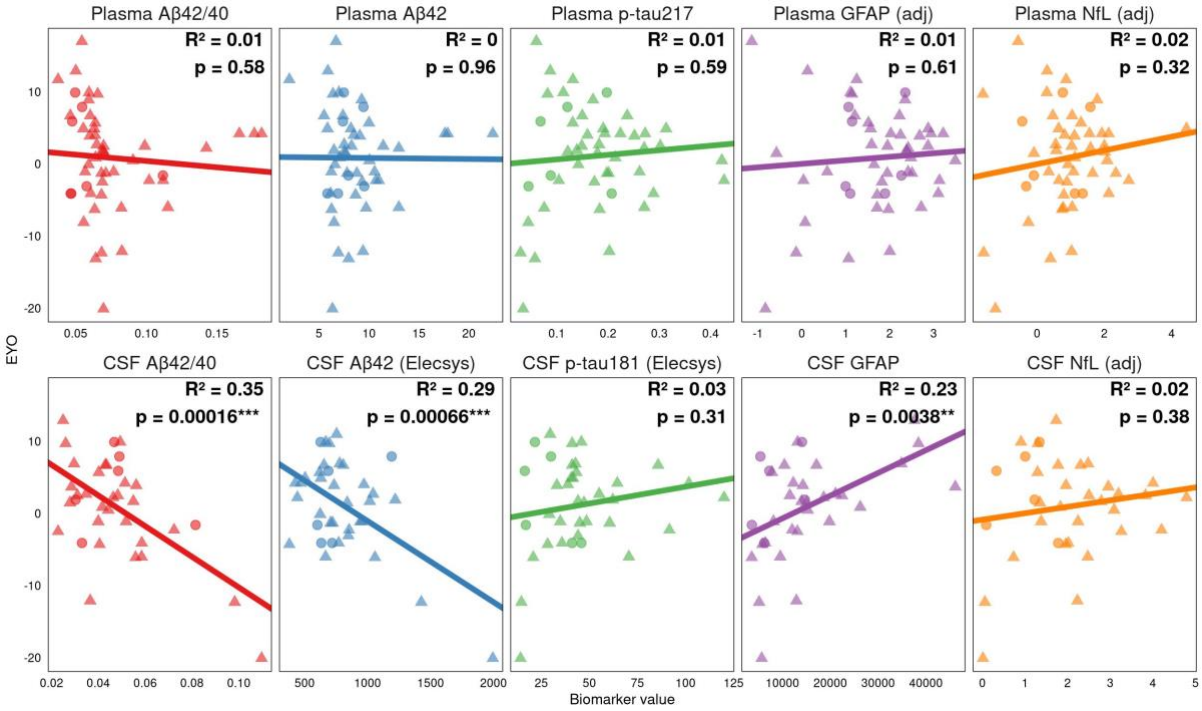
H) Posterior cortical atrophy (PCA)



I) Presence of symptoms



J) Estimated years to symptom onset (EYO)



Supplementary Figure 4 Exploratory analysis of fluid biomarkers and their association with the REVEL score

